

Designing and Building a Computer Science Portfolio

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Abstract

This report documents the design and implementation of a computer-science portfolio created to support a working-student (*Werkstudent*) application in Berlin. The portfolio consists of a website hosted on GitHub Pages, a one-page CV produced in L^AT_EX, and this report. The document explains the structure of the website and its repository, justifies the main design decisions against current evidence on technical résumés and developer portfolios, lists the tools and technologies used, and reflects on the portfolio’s strengths, weaknesses, and planned improvements.

1 Introduction

The portfolio is published at <https://sw3do.github.io> and its source lives at github.com/sw3do/sw3do.github.io. Its purpose is to present, to a hiring manager, concrete evidence of software-engineering ability for a working-student role in Berlin’s international technology sector.

The central design constraint is attention. Recruiters and hiring managers typically spend well under a minute on a first pass over a candidate’s materials (Pesto, 2025), so every artefact is built to communicate competence quickly and to reward a closer look without demanding one. The remainder of this report describes the structure of the website and repository (Section 2), justifies the major design decisions with reference to current practice (Section 3), lists the tools and technologies used (Section 4), and reflects on what works, what does not, and what could be added next (Section 5).

2 Portfolio and Repository Structure

2.1 The website

The site is a single, scrollable page with supporting sub-pages. The landing page is divided into clearly labelled sections — *Hero*, *About*, *Skills*, *Projects*, *Experience* and *Contact* — so a reader can locate any kind of information at a glance. The hero introduces the candidate with a short statement of focus and a terminal-style panel summarising the technology stack, reinforcing the systems-oriented theme of the work.

Two kinds of supporting page add depth without cluttering the landing page. **Case studies** (for example, `/projects/packet-sniffer` and `/projects/jikan-rs`) each follow the same honest structure: the problem, the approach and why it was chosen, a concrete challenge encountered, the outcome, and what could be improved next. A small **blog** hosts short technical notes drawn from the same projects. Only public repositories are featured, so that every repository and demo link a visitor follows resolves correctly.

2.2 The repository

The repository is organised so that files which change together live together:

```
sw3do.github.io/  
  src/  
    components/ layouts/ pages/  
    content/ (projects + blog, as Markdown)  
    data/ (profile, skills, projects)  
    styles/ (design tokens + global CSS)  
  public/ (CV PDF, OG image, favicon, robots.txt)  
  cv/ (LaTeX source + Mert-Celik-CV.pdf)  
  report/ (this report)  
  .github/workflows/ (build + deploy)  
  README.md
```

The site is deliberately **data-driven**: the candidate’s profile, skill groups and project list live in typed data files, and case studies and blog posts are Markdown entries in Astro *content collections*. Presentation components read from these sources, so updating content never means editing layout code. This separation keeps each file focused and the project easy to extend.

3 Design Decisions and Justification

3.1 One coherent brand

The CV, website and this report share a single restrained visual identity: one teal accent colour, a small set of typefaces, and a minimal, generously spaced layout. A consistent identity across artefacts helps a reader perceive them as one professional body of work, and minimalism keeps the most relevant information above the fold where a time-pressed reader will see it ([Scrimba, 2025](#); [Pesto, 2025](#)).

3.2 A one-page, single-column, ATS-safe CV

The CV is a single page in one column. For an early-career candidate this is the strongly recommended format: it forces ruthless curation and survives the automated applicant-tracking systems (ATS) that scramble multi-column layouts ([Tech Interview Handbook, 2025](#); [Job Shinobi, 2026](#)). The structure leads with skills and projects rather than employment history, because for a student the public project portfolio *is* the experience ([University of Washington, 2025](#)). Bullet points follow an action–technology–result shape rather than listing duties ([IGotAnOffer, 2025](#)), and the document is compiled with pdfLaTeX using a glyphtounicode mapping so the exported PDF yields clean, selectable text ([Gutierrez, 2024](#); [ATS Resume AI, 2026](#)).

3.3 An anonymised CV for the Berlin market

The CV omits a photograph, date of birth and home address. Although a photo remains common in German applications, the anonymised CV is an accepted and growing norm, particularly in Berlin’s international, English-speaking technology scene, and it reduces the scope for unconscious bias ([CVCompose, 2026](#)). Language ability is given on the CEFR scale, and German is listed honestly at A1 with a note that it is actively improving — a signal of commitment that German employers value ([Goethe-Institut, 2024](#); [Study-Abroad.org, 2026](#)).

3.4 A static site built with Astro

The website is a static site generated with Astro and deployed through GitHub Actions. Astro ships almost no client-side JavaScript by default, which directly supports the performance and

accessibility goals below, and its content collections make the case studies and blog easy to author and validate ([Astro, 2026](#)). A static build is also the natural fit for GitHub Pages and gives a reproducible, automated deployment pipeline.

3.5 Accessibility, performance and SEO

The site is built on semantic HTML with a correct heading hierarchy, descriptive labels and visible keyboard focus, and it targets WCAG AA colour contrast in both its dark and light themes. These choices improve accessibility and, because the same signals feed Google’s Core Web Vitals, also improve search and perceived quality ([Siteimprove, 2025](#); [OWDT, 2025](#)). The layout is mobile-first, reflecting mobile-first indexing and the reality that many reviewers open links on a phone ([EngageCoders, 2025](#)). Because GitHub Pages cannot run server-side logic, search-engine and social metadata are baked in at build time: per-page titles and descriptions, Open Graph and Twitter cards, JSON-LD structured data, a sitemap and a `robots.txt` ([JekyllPad, 2025](#)).

3.6 Honesty over polish

Every featured project is real, and each case study names a concrete challenge and an honest “what I would improve next”. Vague, metric-free claims and rule-of-three filler are precisely the tells that mark a portfolio as inflated or machine-generated ([Learnist, 2026](#)), so the content favours specific, verifiable statements over decoration.

4 Tools and Technologies

- **Astro + TypeScript** — static site generation with typed data and content collections.
- **CSS custom properties** — a single source of truth for the dark/light design tokens.
- **astro-icon** with **Lucide** and **Simple Icons** — accessible, build-time inlined SVG icons.
- **@fontsource** (Space Grotesk, Inter, JetBrains Mono) — self-hosted fonts for performance and offline builds.
- **@astrojs/sitemap** — automatic sitemap generation.
- **GitHub Pages + GitHub Actions** — hosting and continuous deployment on every push.
- **Git** — version control.
- **L^AT_EX** (pdfLaTeX, lmodern, natbib, glyphtounicode) — the CV and this report.
- **Pillow** — programmatic generation of the social-share (Open Graph) image.
- **Google Lighthouse** — auditing performance, accessibility, best practices and SEO.

The toolchain was deliberately kept lightweight and portable: the documents compile with a minimal L^AT_EX installation and no external citation processor, and the site builds with a single `npm` command.

5 Reflection

5.1 Strengths

The portfolio shows genuine breadth and depth: production-style work across Rust, Go, Python and TypeScript, spanning a published library, a systems/networking tool, a machine-learning service and a cross-platform desktop application. Quality is measurable rather than asserted —

the site is accessible, fast and search-optimised — and the CV, website and report read as one coherent identity. The deployment is reproducible through continuous integration, and the CV is verifiably ATS-safe.

5.2 Weaknesses

Formal professional experience is still limited, and several featured projects are libraries or command-line tools without a hosted live demo, which is less immediately persuasive than an interactive web app. German is only at A1. The site has no automated test suite, and as a single-author project it has not had external code review.

5.3 Future work

Planned improvements include: per-project quantitative metrics (benchmarks, test coverage, download counts); additional case studies; a custom domain; automated link-checking and Lighthouse runs in continuous integration to prevent regressions; internationalisation (English, German and Turkish); and a more active blog. Adding hosted demos for the library and CLI projects — for instance an interactive playground — would also strengthen the first impression.

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A Lighthouse Audit

A Google Lighthouse audit of the deployed site is included here, reporting the Performance, Accessibility, Best Practices and SEO scores.